

① Stars Patterns

```
public class pattern {
```

```
    public static void main (String[] args)
```

```
    {
```

```
        for (int i=1; i<=5; i++)
```

```
        {
```

```
            for (int j=1; j<=5; j++)
```

```
            {
```

```
                System.out.print("*");
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

Square

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Rows - (1 to 5)

cols - (1 to i)

Pattern.java

```
1
2 public class Pattern {
3     public static void main (String[] args){
4
5         for(int i = 1; i<=5; i++)
6         {
7             for(int j =1; j<=5; j++)
8             {
9                 System.out.print("*");
10            }
11            System.out.println();
12        }
13    }
14 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop\java"
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Hollow square

```
*****
*       *
*       *
*       *
*****
```

```
public class pattern {
    public static void hollow_Square (int n, int m)
    {
        for (int i = 1; i <= n; i++)
        {
            for (int j = 1; j <= m; j++)
            {
                if (i == 1 || i == n || j == 1 || j == m)
                {
                    System.out.print("*");
                }
                else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }

    public static void main (String args[]) {
        hollow_Square (5,5);
    }
}
```

Pattern.java

```
1
2 public class Pattern {
3     public static void hollow_rectangle( int n , int m)
4     {
5         for(int i = 1; i<=n; i++)
6         {
7
8             for(int j =1; j<=m; j++)
9             {
10                if(i==1 || i==n || j==1 || j==m){
11                    System.out.print("*");
12                }
13                else{
14                    System.out.print(" ");
15                }
16            }
17            System.out.println( );
18        }
19    }
20 }
21
22 public static void main (String args[]){
23     hollow_rectangle(5,5);
24 }
25
26
27 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

P5 C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish"

* *

* *

* *

P5 C:\Users\Aashish\OneDrive\Desktop\java>

Triangle -

```
public class pattern{
```

```
    public static void main (String args[])
```

```
    {
```

```
        for(int i = 1; i <= 5; i++) {
```

```
            for(int j = 1; j <= i; j++)
```

```
            {
```

```
                System.out.print("*");
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

```
*
* *
* * *
* * * *
* * * * *
*
```

```
Click to add a breakpoint
1 public class pattern {
2
3
4     public static void main(String args[]){
5
6         for(int i = 1; i <= 5; i++){
7
8             for(int j = 1; j <= i; j++){
9
10                System.out.print("*");
11            }
12            System.out.println();
13        }
14    }
15 }
16
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop\
*
**
***
****
*****
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Pyramid :-

```
public class Pattern {  
    public static void main (String args[])  
    {  
        for (int i=1; i<=5; i++){  
            for (int j=1; j<=5-i; j++){  
                System.out.print(" ");  
            }  
            for (int j=1; j<=(2*i)-1; j++){  
                System.out.print("*");  
            }  
            System.out.println();  
        }  
    }  
}
```

```
}  
}  
}
```

```
1  
2 public class Pattern {  
3  
4     public static void main(String args[]){  
5  
6         for(int i =1; i<=5; i++){  
7  
8             for(int j = 1; j<=5-i; j++){  
9  
10                System.out.print(" ");  
11            }  
12            for(int j =1; j<=(2*i)-1; j++){  
13                System.out.print("*");  
14            }  
15            System.out.println();  
16        }  
17    }  
18 }  
19
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive  
*  
***  
*****  
*****  
*****  
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Inverted Triangle

```
public class Pattern {  
    public static void main(String args[])  
    {  
        for (int i = 5; i >= 1; i--) {  
            for (int j = 1; j <= 5-i; j++) {  
                System.out.print(" ");  
            }  
            for (int j = 1; j <= (2*i)-1; j++) {  
                System.out.print("*");  
            }  
            System.out.println();  
        }  
    }  
}
```

```
*****  
*****  
*****  
***  
*
```

outer loop (n to 1)

inner loop -

spaces (n-i)

• ((2*i)-1) for *

Pattern.java

```
1  
2 public class Pattern {  
3  
4     public static void main(String args[]) {  
5  
6  
7  
8         for (int i = 5; i >= 1; i--) {  
9  
10  
11             for (int j = 1; j <= 5-i; j++) {  
12  
13                 System.out.print(" ");  
14             }  
15             for (int j = 1; j <= (2*i)-1; j++) {  
16                 System.out.print("*");  
17             }  
18             System.out.println();  
19         }  
20  
21  
22     }  
23 }  
24
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop\java"

```
*****  
*****  
***  
*
```

PS C:\Users\Aashish\OneDrive\Desktop\java>

Diamond

```
public class Pattern {
    public static void main(String args[]) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= (5-i); j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= (2*i)-1; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
        for (int i = 5; i >= 2; i--) {
            for (int j = 1; j <= (5-i); j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= (2*i)-1; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

```

      *
    ***
  *****
*****
*****
*****
  *****
    ***
      *
```

```
1
2 public class Pattern {
3
4     public static void main(String args[]) {
5
6         for (int i = 1; i <= 5; i++) {
7
8             for (int j = 1; j <= 5-i; j++) {
9
10                System.out.print(" ");
11            }
12            for (int j = 1; j <= (2*i)-1; j++) {
13                System.out.print("*");
14            }
15            System.out.println();
16        }
17
18        for (int i = 5; i >= 2; i--) {
19
20            for (int j = 1; j <= 5-i; j++) {
21
22                System.out.print(" ");
23            }
24            for (int j = 1; j <= (2*i)-1; j++) {
25                System.out.print("*");
26            }
27            System.out.println();
28        }
29    }
30 }
31
32
33
34
35
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "C:\Users\Aashish\OneDrive\Desktop\java\" ; if ($?) {
*
***
*****
*****
*****
*****
*****
***
*
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Half Diamond

```
public class Pattern {  
    public static void main(String args[]) {
```

```
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print("*");  
            }
```

```
            System.out.println();  
        }
```

```
        for (int i = 4; i >= 1; i--) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print("*");  
            }
```

```
            System.out.println();  
        }
```

```
    }
```

```
}
```

logic → outer loop (1 to 5)

inner for space (1 to i)

for star (

same it's opposite

only outer loop (n to 1)

```
*  
**  
***  
****  
*****  
****  
***  
**  
*  
*
```

```
1  
2 public class Pattern {  
3  
4     public static void main(String args[]) {  
5  
6  
7  
8         for (int i = 1; i <= 5; i++) {  
9  
10            for (int j = 1; j <= i; j++) {  
11  
12                System.out.print("*");  
13            }  
14  
15            System.out.println();  
16        }  
17        for (int i = 4; i >= 1; i--) {  
18  
19            for (int j = 1; j <= i; j++) {  
20  
21                System.out.print("*");  
22            }  
23  
24            System.out.println();  
25        }  
26    }  
27 }  
28 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDr

*

**

**

*

PS C:\Users\Aashish\OneDrive\Desktop\java>

Square

```
public class Pattern {  
    public static void main(String args[]) {  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= 5; j++) {  
                System.out.print("1");  
            }  
            System.out.println();  
        }  
    }  
}
```

```
11111  
11111  
11111  
11111  
11111
```

```
1  
2 public class Pattern {  
3  
4     public static void main(String args[]) {  
5  
6         for (int i = 1; i <= 5; i++) {  
7  
8             for (int j = 1; j <= 5; j++) {  
9  
10                System.out.print("1");  
11            }  
12            System.out.println();  
13        }  
14    }  
15 }  
16  
17  
18  
19  
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish"  
11111  
11111  
11111  
11111  
11111  
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Incrementing

```
public class Pattern {  
    public static void main(String args[]) {
```

```
        for(int i = 1; i <= 5; i++) {
```

```
            for(int j = 1; j <= 5; j++)
```

```
            {  
                System.out.print(j);
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

```
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5
```

```
Pattern.java  
1  
2 public class Pattern {  
3  
4     public static void main(String args[]) {  
5  
6  
7  
8         for(int i = 1; i <= 5; i++) {  
9  
10            for(int j = 1; j <= 5; j++) {  
11  
12                System.out.print(j);  
13            }  
14  
15            System.out.println();  
16        }  
17    }  
18 }  
19 }  
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive  
12345  
12345  
12345  
12345  
12345  
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Right Triangle

```
public class Pattern{  
    public static void main (String args[])  
    {  
        for (int i=1; i<=5; i++){  
            for (int j=1; j<=i; j++){  
                System.out.print (j);  
            }  
            System.out.println();  
        }  
    }  
}
```

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

Pattern.java

```
1  
2 public class Pattern {  
3  
4     public static void main(String args[]){  
5  
6  
7  
8         for(int i =1; i<=5; i++){  
9  
10            for(int j = 1; j<=i; j++ ){  
11  
12                System.out.print(j);  
13            }  
14  
15            System.out.println();  
16        }  
17    }  
18 }  
19 }  
20
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish"
1
12
123
1234
12345
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Incrementing Diamond Inverted Pattern

```
public class Pattern {
    public static void main(String args[]) {
        for (int i=1; i<=5; i++){
            for (int j=1; j<=i; j++){
                int n=i+2;
                System.out.print(n);
            }
            System.out.println();
        }
        for (int i=4; i>=1; i--){
            for (int j=1; j<=i; j++){
                int n=i+2;
                System.out.print(n);
            }
            System.out.println();
        }
    }
}
```

```
3
44
555
6666
77777
6666
555
44
3
```

```
1 C:\Users\Aashish\OneDrive\Desktop\java\Pattern.java
2 public class Pattern {
3
4     public static void main(String args[]) {
5
6         for(int i= 1; i<=5; i++)
7         {
8             for(int j =1; j<=i; j++){
9
10                int n =i+2;
11
12                System.out.print(n);
13            }
14            System.out.println();
15        }
16
17        for(int i= 4; i>=1; i--)
18        {
19            for(int j =1; j<=i; j++){
20
21                int n =i+2;
22
23                System.out.print(n);
24            }
25            System.out.println();
26        }
27
28    }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\
3
44
555
6666
77777
6666
555
44
3
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Sandwich Pattern

```

public class Pattern{
    public static void main(String args[]){
        for( int i=3; i>=1; i--){
            for( int j=1; j<=3-i; j++){
                System.out.print(" ");
            }
            for( int k=1; k<=(2*i)-1; k++){
                int n = 6-i;
                System.out.print(n);
            }
            System.out.println();
        }
        for( int i = 1; i<=2; i++){
            for( int j=1; j<=2-i; j++){
                System.out.print(" ");
            }
            for( int k = 1; k<=(2*i)+1; k++){
                int n = 7-(2+i);
                System.out.print(n);
            }
            System.out.println();
        }
    }
}

```

```

33333
444
5
444
33333

```

```

1
2 public class Pattern {
3
4     public static void main(String args[]){
5
6         for(int i= 3; i>=1; i--){
7             {
8                 for(int j =1; j<=3-i; j++){
9                     System.out.print(" ");
10                }
11                for(int k=1; k<=(2*i)-1; k++){
12                    int n =6-i;
13                    System.out.print(n);
14                }
15                System.out.println();
16            }
17
18            for(int i= 1; i<=2; i++){
19                {
20                    for(int j =1; j<=2-i; j++){
21                        System.out.print(" ");
22                    }
23                    for(int k=1; k<=(2*i)+1; k++){
24                        int n =7-(2+i);
25                        System.out.print(n);
26                    }
27                    System.out.println();
28                }
29            }
30        }
31    }
32 }
33
34
35
36 }
37
38 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop\java"
33333
444
5
444
33333
PS C:\Users\Aashish\OneDrive\Desktop\java>

```